Notes for the First Lecture

"Imagine now that each player $k = 1, \ldots, n$, *instead of making each decision as the necessity for it arises, makes up his mind in advance for all possible contingencies; i.e., that the player k begins to play with a complete plan: a plan which specifies what choices he will make in every possible situation, for every possible actual information which he may possess at that moment [. . .] We call such a plan a strategy. Observe that if we require each player to start the game with a complete plan of this kind, i.e., with a strategy, we by no means restrict his freedom of action. In particular, we do not thereby force him to make decisions on the basis of less information than there would be available for him in each practical instance in an actual play.*
The player k must choose his strategy [. . .] without information concerning the choices of the other players, or of the chance events (the umpire's choice). This must be so since all the information he can at any time possess is already embodied in his strategy."
— von Neumann and Morgenstern (1944)

"[. . .] It is clear that for the purpose of the investigation of the problem of perfectness, the normal form is an inadequate representation of the extensive form."
— Selten (1975)

"Analyses that ignore the role of beliefs, such as analysis based on normal form representation, inherently ignore the role of anticipated actions off the equilibrium path."
— Kreps and Wilson (1982)

These notes follow the classical point of view regarding game theory, starting with von Neumann and Morgenstern and Nash: it is a *prescriptive* theory, attempting to describe how rational players should act. A representation of a game already encompasses every aspect of the real situation: communication possibilities, commitment opportunities, etc.

This is a debatable perspective: we might be interested in solution concepts that are descriptive (evolutionary game theory, behavioral economics, social psychology, learning, etc.); we might and often do define the solution concept in a way that already encapsulates communication opportunities, commitment, etc.

But this is the point of view we will adopt today. It follows that we take as granted that: behavior should be *self-enforcing*; all data is *common knowledge* so that any conclusion leading player i to follow a given behavior is a conclusion that player j can also draw. The italicized words are taken as self-evident here. Their content will be revisited in future lectures. For now, we take for granted that they imply that whatever solution concept we select must be within the set of Nash equilibria.

Throughout, we also assume Savage's postulates, namely, that players' preferences can be represented by a utility function and (subjective) beliefs. In the context of today's lecture, we adopt the stronger vNM expected utility framework. I recommend Kreps' *Notes on the Theory of Choice* for background reading on decision theory under uncertainty.

Games can be represented in many ways. The fundamental way of representing games is via the extensive form. Following Kuhn (1952), this is usually done using a labeled graph.

1 Brief Reminders: Extensive and Normal Form

What we need to know is:

1. The set of players.
2. The order of moves: who moves when.
3. What the players' choices are when they move.

4. What each player knows when she makes her choices.
5. The probability distribution over exogenous events.
6. The players' payoffs as a functions of all the choices and events.

The penultimate point needs some additional explanation. Consider, for instance, playing at a slot-machine. It makes sense to model this as a game with only one player, the gambler. But her payoff depends also on something else, namely chance, often referred to as **Nature** in game theory, or player 0. Nature is a very special, additional, player, without preferences, and whose way of playing must be specified to begin with. Hence, we must specify the probability distribution over exogenous events.

All this information is provided by the extensive form. Recall that a **partition** of a finite set B is a collection of disjoint subsets of B whose union is B.

Definition 1 *An n-**person game in extensive form** is a tuple $\Gamma := (T, P, H, C, p, u)$ where:*

1. *T is a finite tree, with nodes X and edges E.*
2. ***The player partition:** $P := \{P_0, P_1, \ldots, P_n\}$ is a partition of $X \setminus Z$; this indicates, for each decision node, whose turn it is to play.*
3. ***The information partition:** $H = \{H_1, \ldots, H_n\}$, where H_i is a partition of P_i, for $i = 1, \ldots, n$, such that no path in T intersects some element $h \in H_i$ twice, and if $x, x' \in h \in H_i$, then x and x' have the same number of children; this indicates decision nodes that a player cannot distinguish.*
4. ***The choice partition:** C is a partition of E such that, $\forall i = 1, \ldots, n$, if $(x, x'') \in c \in C$, with $x \in h \in H_i$, then $\forall x' \in h$, $\exists! x'''$ such that $(x', x''') \in c$; these are the choices available to the players.*
5. ***The probability assignment:** $p := \{p_x : x \in P_0\}$, where p_x is a probability distribution over the children of x; this indicates how Nature plays at each of its decision nodes.*

6. ***The utilities:*** *$u := \{u_1, \ldots, u_n\}$, where $u_i : Z \to \mathbf{R}$; this indicates, for each terminal node, the payoff of each player.*

Choices are also called **actions**, or **moves**, and often denoted a_i (for actions of Player i). We write C_h, or $A(h)$, for the choices available at $h \in H_i$.

The last item is often replaced by a **rational preference relation** $\succsim_i$ for each Player i (a complete, transitive binary relation) over terminal nodes: if $z, z' \in Z$, $z \succsim_i z'$ means that i prefers z to z' (weakly). We write $\succ_i$ for the strict version, *i.e.*, $z \succ_i z'$ if $z \succsim_i z'$, but not $z' \succsim_i z$. We write $z \sim_i z'$ if both $z \succsim_i z'$ and $z' \succsim_i z$, so that Player i is indifferent over z and z'. Rational preference relations are in a sense more general than utilities (every utility vector defines a preference relation, but not conversely), and we shall work with those as long as possible.

Almost all games in the literature (and all of those we will talk about) are games with perfect recall. No player ever forgets what she once knew, and what she once did. We have already required that no path intersects the same information set twice, but this is not enough. Formally:

Definition 2 *A game has **perfect recall** if whenever $x', x'' \in h \in H_i$, and $x \in P_i$ is an ancestor of x', then $\exists \hat{x} \in h' \in H_i$, such that (i) $x \in h'$, (ii) $\hat{x}$ is an ancestor of x'', (iii) the choice at $\hat{x}$ on the path $(\hat{x}, x'')$ is the same as the choice at x on the path (x, x').*

Intuitively, if Player i cannot distinguish two nodes, she must have played the same way to reach each of them. Henceforth, we also implicitly assume that the games we are dealing with are games with perfect recall.

An important class of games are games of perfect information.

Definition 3 *An extensive game Γ is a game with **perfect information** if, $\forall i = 1, \ldots, n$, $\forall h \in H_i$, h is a singleton.*

An extensive form game without perfect information is said to have **imperfect information**.

Checkers and chess have perfect information. Bridge and poker do not (and they also have chance). Matching pennies has no chance, but no perfect information either.

1.1 What's a Strategy?

We have now described what players can do. Let us talk about what they plan to do. We must introduce here the notion of a strategy. A strategy for a player is a plan of action that is complete and respects information sets. Let $A_i := \cup_{h \in H_i} A(h)$.

Definition 4 *A* ***pure strategy*** s_i *for Player* i *is a map* $s_i : H_i \to A_i$ *such that* $\forall h \in H_i$, $s_i(h) \in A(h)$.

The set of all strategies for Player i in a given game Γ is denoted $S_i := \times_{h \in H_i} A(h)$. A **pure strategy profile** is a vector of strategies $s := (s_1, \ldots, s_n)$. Let $S := \times_i S_i$.

Note that a strategy defines what a player does even at information sets that are precluded by what her strategy specifies at earlier information sets.

For reasons that will become clear later on, it is also important to let players make unpredictable, *i.e.*, random choices (think about poker). There are two ways to define such random choices, called **mixed** and **behavior** strategies. We will come back to the first one later. The second one is defined as follows. Given any finite set B, let $\triangle B$ define the probability distributions (also called lotteries) over B. More formally, if $|B|$ denotes the number of elements in B, then $\triangle B := \{p \in \mathbf{R}^{|B|} | \sum_{j \in B} p_j = 1, \text{ and } p_j \geq 0, \text{ all } j\}$.

Definition 5 *A* ***behavior strategy*** *for Player* i *is a map* $\sigma_i : H_i \to \triangle A_i$ *such that* $\forall h \in H_i$, $\sigma_i(h) \in \triangle A(h)$.

So it is simply an element of $\Sigma_i := \times_{h \in H_i} \triangle A(h)$. A behavior strategy specifies a probability distribution at each information set, and the distributions at different information sets are independent. Observe that a pure strategy is just a special case of a behavior strategy. A behavior strategy profile is a collection $\sigma := (\sigma_1, \ldots, \sigma_n)$, and let $\Sigma := \times_i \Sigma_i$. Any behavior strategy profile generates recursively a probability distribution over terminal nodes Z.

Observe that, if Nature is not a player, we could immediately have defined the game by specifying simply the sets S_i, and functions $u_i : S \to \mathbf{R}$, $i = 1, \ldots, n$, by defining $u_i(s)$ as the payoff obtained from the terminal node that is reached under the pure strategy profile s. (Even if Nature was a player, we could do so, by computing the relevant expectations.) This leads to the following definition.

Definition 6 *A* ***normal-form game*** *is a collection* $(S_i, u_i)_{i=1,\ldots,n}$*, where* S_i *is a finite set and* $u_i : \times_i S_i \to \mathbf{R}$*.*

We write henceforth for short (S, u) for the normal form, where $S := \times_i S_i, u := (u_1, \ldots, u_n)$. Again, a more general version of the definition uses preference relations $\succsim_i$, all i, instead of utility functions.

In general, in a given extensive form, there are many strategies that are equivalent, in the sense that, no matter the strategy profile used by the other players, those strategies induce the same distribution over terminal nodes $z \in Z$. For instance, this is often the case if a given player gets to move twice along some path: suppose that Player i chooses first between a_i and a_i', and later, if she chooses a_i, she must choose between b_i and b_i'. If her strategy calls for a_i' at the first information set, then what she would do at the second had she chosen a_i instead is irrelevant to the outcome of the game, given the strategies of the other players. (Nevertheless, it is important when we solve extensive form games to specify strategies fully, because whether the strategies of the other players are optimal, in the sense to be defined later, can only be evaluated relative to the entire strategy of Player i.) To simplify the analysis in the normal form, we sometimes eliminate all but one of these equivalent strategies, which leads to the following concept.

Definition 7 *The* ***reduced normal form*** *of an extensive-form game is obtained by identifying equivalent pure strategies (i.e., by eliminating all but one member of each equivalence class).*

The reduced normal form can be further simplified:

Definition 8 *The* ***fully reduced normal form*** *of an extensive-form game is obtained by eliminating all pure strategies that can be represented as convex combinations of other pure strategies.*

Alternatively, for a given extensive form, we might consider a more general normal form.

Definition 9 *The **agent normal form** of an extensive-form game is obtained by taking the normal form of that extensive form, identifying each information set of the extensive form as a different player (all with the same payoff).*

Observe that there is another way of defining random choices: a player might choose a probability distribution over S_i, that is an element $l_i \in \triangle S_i$. This is a more complicated object. Formally:

Definition 10 *A **mixed strategy** for Player i is a probability distribution over S_i, that is, an element $l_i \in \triangle S_i$.*

Let L_i denote the set of all mixed strategies in a given game Γ in extensive form. Clearly, $\Sigma_i \subset L_i$, so that the set of mixed strategies is larger than the set of behavior strategies. It is then natural to ask whether these two ways of defining random choices are equivalent. We first need to be precise about what equivalent means.

Definition 11 *Let Γ be an extensive form game. The strategies $\sigma_i \in \Sigma_i$ and $l_i \in L_i$ are **realization-equivalent** if, for every profile $l_{-i} \in L_{-i}$, (σ_i, l_{-i}) and (l_i, l_{-i}) induce the same distribution over terminal nodes Z.*[1]

Since $\Sigma_i \subset L_i$, for every $\sigma_i \in \Sigma_i$, we can find a mixed strategy l_i such that l_i are σ_i are realization-equivalent. The converse is not obvious, and only holds if the game has perfect recall. We omit the proof.

Theorem 1 ***(Kuhn theorem)*** *Let Γ be an extensive form game of perfect recall. Let $l_i \in L_i$ be a mixed strategy. Then there is $\sigma_i \in \Sigma_i$ such that l_i and σ_i are realization-equivalent.*

Because of this theorem, and our focus on games with perfect recall, we need not worry about the distinction. We shall typically use behavior strategies, $\sigma_i \in \Sigma_i$, since they are simpler, but will typically refer to them as mixed strategies.

[1]The notation $-i$ refers to all players but i; *e.g.*, if $n = 4$, $-2 = \{1, 3, 4\}$.

2 Desiderata

2.1 Invariance

An important question that has “plagued” game theorists is the following:

Is there no loss of generality in focusing on the normal form?

This question is motivated by the fact that the normal form is simpler to analyze (in particular, we can use linear algebra). What makes the question tricky is that it is difficult to answer this question without also adding: *with what solution concept?* Indeed, the naive answer one usually receives is: the extensive form is more appropriate, since subgame-perfection rules out Nash equilibria of the normal form that rely on non-credible threats. But this is ignoring that more restrictive solution concepts than Nash equilibrium can be defined in the normal form –or, for that matter, than subgame perfect Nash equilibrium in the extensive form.

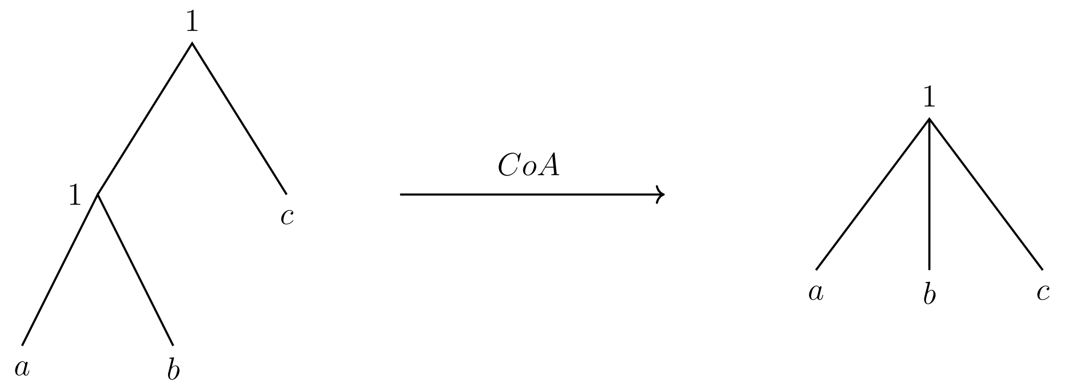
The early literature does not associate a solution concept to this question, and asks whether we can define what it takes for two extensive forms to share the same normal form, and whether the differences between any two such extensive forms are “reasonable,” that is, make no difference from a strategic point of view.

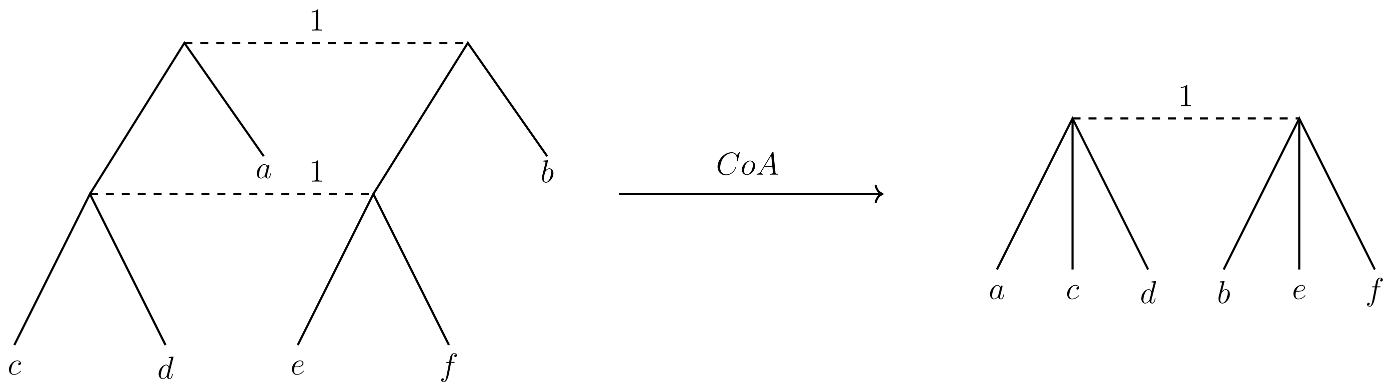
What is the relationship between any two extensive forms that share the same <u>reduced</u> normal form?

The answer, in the case of games with no move by Nature, is provided by the following transformations. Any two extensive forms of perfect recall (without moves by Nature) that share the same normal form differ by some sequence of the following transformations. (Conversely, any such transformation preserves the reduced normal form.)

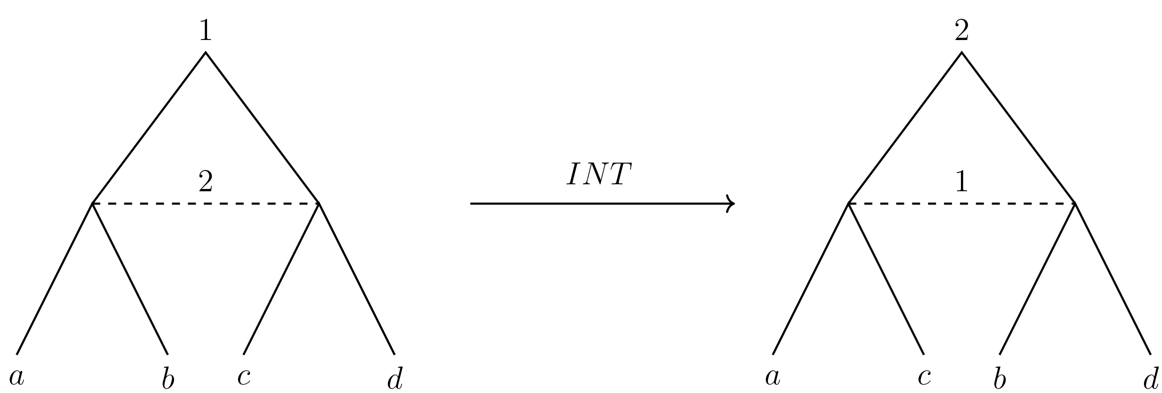
Here are the so-called Thompson-Elmes-Reny transformations. Each transformation is illustrated by a “trivial” example, and then a more sophisticated one. Formal definitions are ignored.

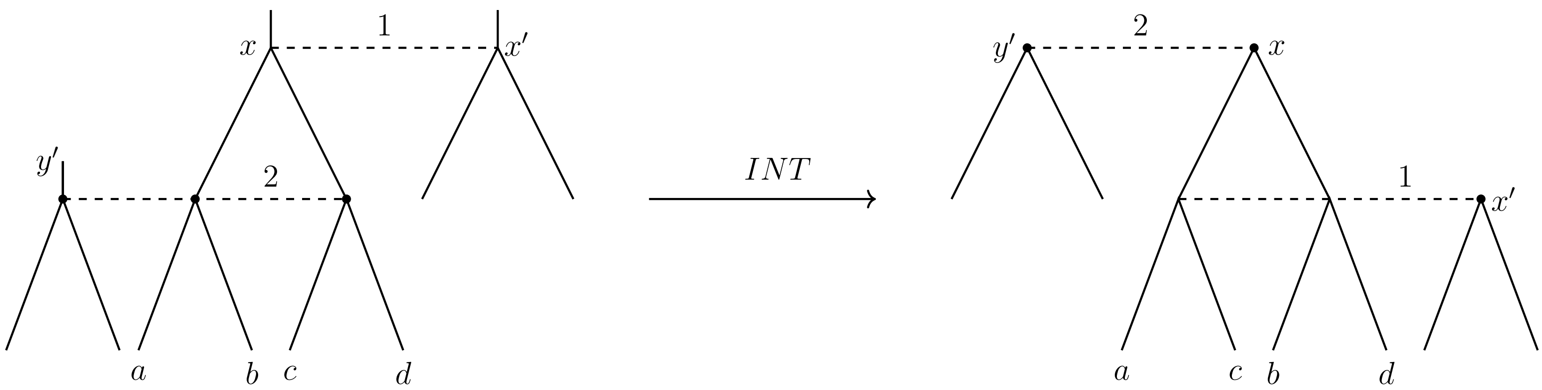
First, coalescing:





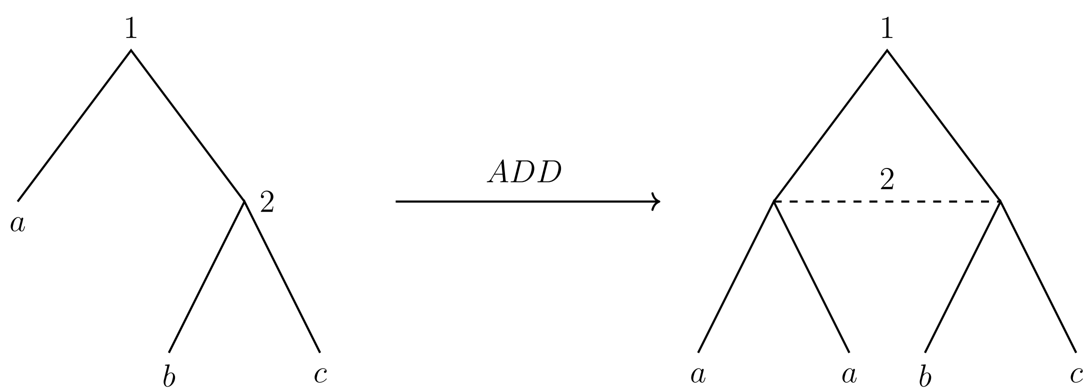
Second, interchanging:

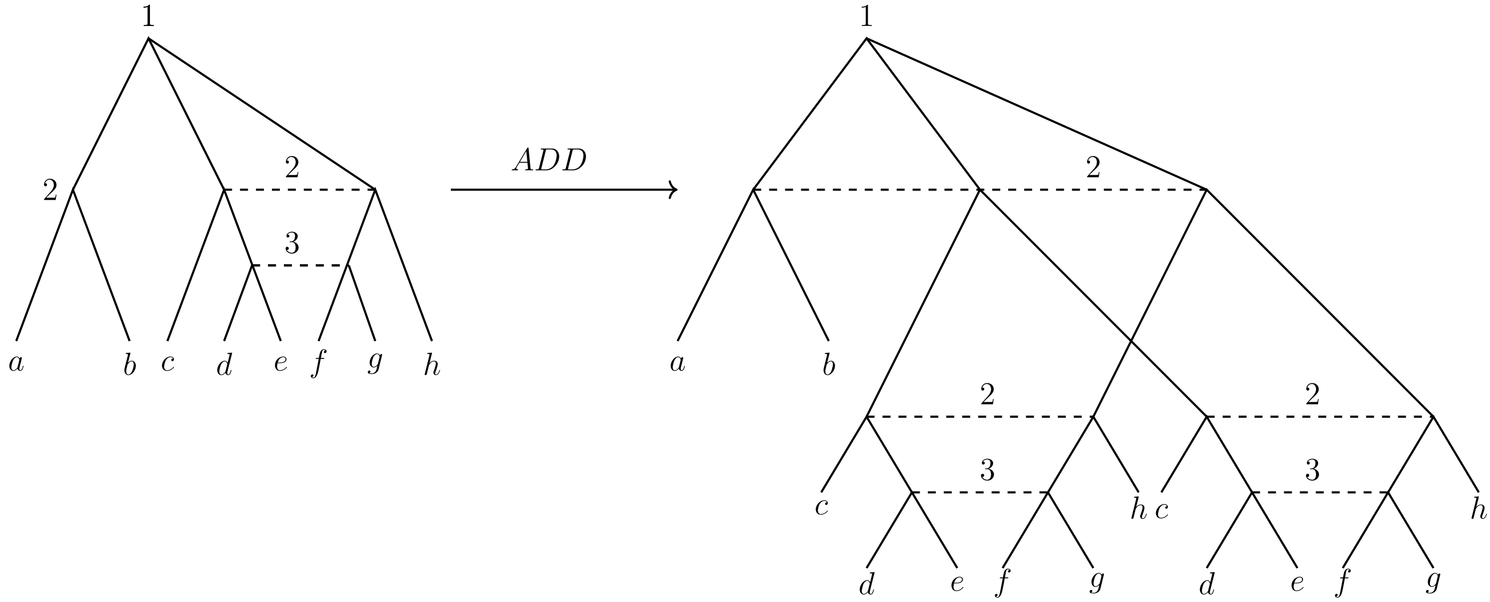




We note that we certainly want to view interchange as innocuous, since simultaneous moves are necessarily represented as sequential moves in a game tree.

Third, adding:





Kohlberg and Mertens (1986) note that this can readily be extended to games with moves by Nature by assuming that those three transformations also apply to Nature, treating Nature as a player; and assuming additionally that, whenever a move by Nature leads only to terminal nodes, it is equivalent to the extensive form in which this move is replaced with the corresponding expected payoff.

Kohlberg and Mertens further introduce a final transformation, namely, if an information set is followed by no other information set, one may add or delete actions at that information set that are convex combinations of other actions at that information set. With this additional transformation, any two extensive-form games have the same fully reduced normal form if, and only if, they differ via a sequence of these five transformations.

Of course, if we take for granted that these five transformations are inessential, then the solution concepts we adopt should treat any two extensive-form games with the same fully reduced normal form equivalently. This is called *invariance*. Let us take subgame-perfect Nash equilibrium (SPNE), to begin with, and consider the following two games illustrated in Figures 1 and 2.

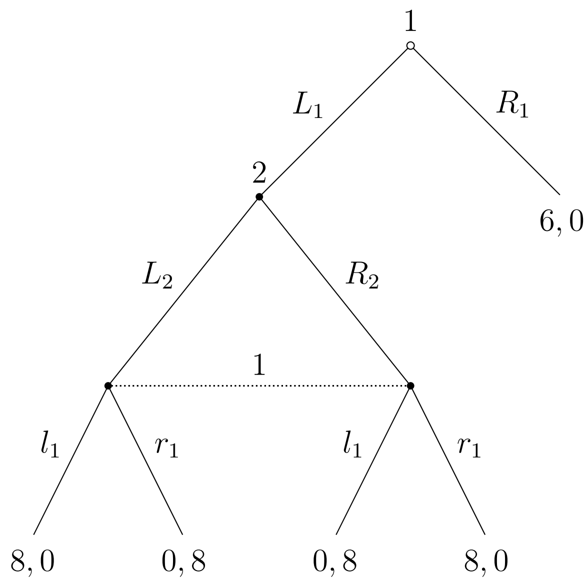


Figure 1: What's the SPNE of this game?

Each of these two games has a unique proper subgame, and it readily follows that they each admit only one SPNE. In the first, Player 2 must randomize $(1/2) - (1/2)$; in the second, she must play L_2 with probability $5/12$, and R_2 with probability $7/12$. (In both cases, Player 1 must choose R_1 at the first node; she must play r_1 with probability $1/3$, and m_1 with probability $2/3$ at the second information set, in the second game.)

We note that the two games have the reduced normal-form representations in Figures 3 and 4.

So, clearly, they share the same fully reduced normal form.

Hence, SPNE does not satisfy invariance. When picking a solution concept, we must either give up sequential rationality (or even simply, that a solution of a game induces a solution in any subgame), or give up invariance, at least relative to the fully reduced normal form. Following this route is disturbing, because it means that one of the transformations isn't innocuous after all.

In fact, sequential equilibrium –the usual solution concept invoked to capture sequential rationality– does not satisfy invariance with respect to *coalesce*.

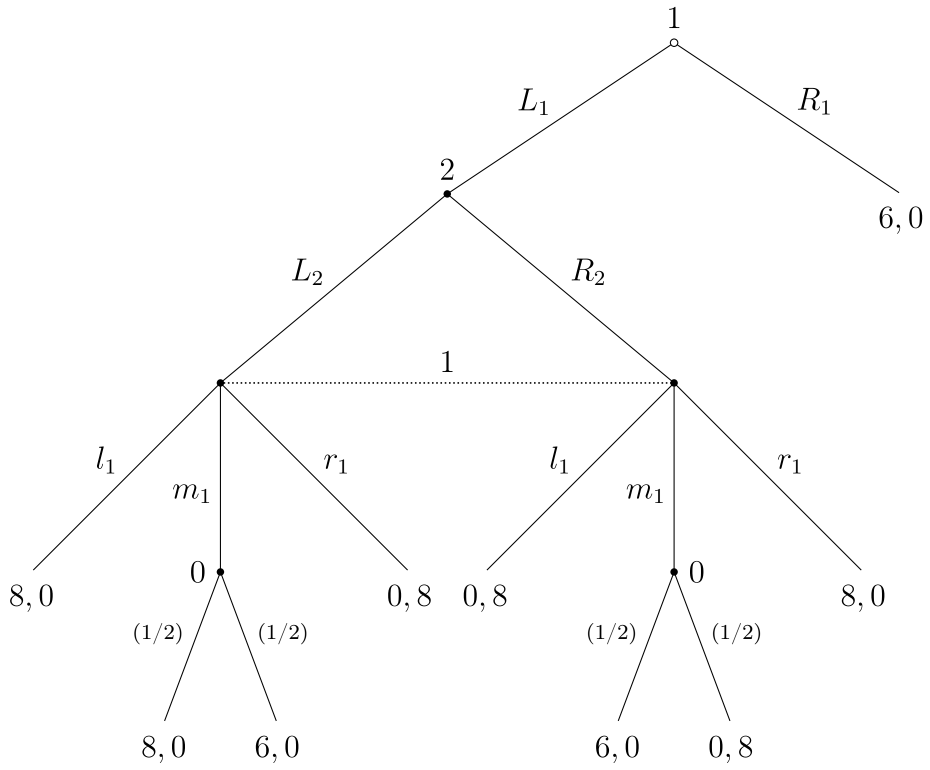


Figure 2: What's the SPNE of this game?

	L_2	R_2
R_1	6, 0	6, 0
$L_1 l_1$	8, 0	0, 8
$L_1 r_1$	0, 8	8, 0

Figure 3: Reduced Normal Form of Figure 1

	L_2	R_2
R_1	$6,0$	$6,0$
$L_1 l_1$	$8,0$	$0,8$
$L_1 r_1$	$0,8$	$8,0$
$L_1 m_1$	$7,0$	$3,4$

Figure 4: Reduced Normal Form of Figure 2

First, some reminders:

Reminders: Fix throughout some extensive form game Γ.

Definition 12 *A **system of beliefs** is a function $\mu: X \setminus Z \to [0,1]$, such that for each information set h, $\sum_{x \in h} \mu(x) = 1$.*

For each node $x \in h$, $\mu(x)$ represents the probability of being at node x, conditional on being at information set h. (We sometimes write $\mu(\cdot \mid h)$ to emphasize the conditioning event.)

Definition 13 *An **assessment** is a pair (σ, μ) where $\sigma \in \Sigma$ is a (behavior) strategy profile, and μ is a system of beliefs.*

The set of all assessments is denoted Ψ. Remember that player i's payoff is determined by the payoff function $u_i : Z \to \mathbf{R}$. Given an assessment $(\sigma, \mu) \in \Psi$, and an information set $h \in H_i$, we can compute the distribution over terminal nodes $z \in Z$, conditional on having reached h. Let $u_i(\sigma \mid h, \mu)$ denote the expected utility of Player i, given that information set h is reached, that the players' beliefs are given by μ, and that the strategies are σ.

Definition 14 *An assessment $(\sigma, \mu) \in \Psi$ is **sequentially rational** if, $\forall i, \forall h \in H_i$, $\forall \sigma_i' \in \Sigma_i$,*

$$u_i(\sigma \mid h, \mu) \geq u_i((\sigma_i', \sigma_{-i}) \mid h, \mu).$$

Given the set of strategy profiles Σ, let Σ^0 denote the strategy profiles such that every player assigns positive probability to all pure strategies. Given $\sigma \in \Sigma^0$, the probability $\mathbf{P}^\sigma(x)$ is strictly positive for all nodes $x \in X$, so that Bayes' rule pins down beliefs at each information set: $\mu(x) = \mathbf{P}^\sigma(x)/\mathbf{P}^\sigma(h)$ for all $x \in h$. Let also Ψ^0 denote the set of all assessments $(\sigma, \mu) \in \Psi$ such that $\sigma \in \Sigma^0$ and μ is (uniquely) derived from σ according to Bayes' rule.

Definition 15 *An assessment $(\sigma, \mu) \in \Psi$ is **consistent** if, for some $\{(\sigma^n, \mu^n)\}_{n\in\mathbf{N}} \subset \Psi^0$,*

$$(\sigma, \mu) = \lim_{n\to\infty} (\sigma^n, \mu^n).$$

Note that the strategy profile σ need not be in Σ^0. But it must be regarded as the limit of some sequence of elements in Σ^0, and the belief μ must be the limit of the corresponding beliefs.

Definition 16 *A **sequential equilibrium** is an assessment $(\sigma, \mu) \in \Psi$ that is sequentially rational and consistent.*

It is not hard to see the following. The proofs of the next two results are omitted.

Proposition 1 *If (σ, μ) is a sequential equilibrium, then σ is a subgame-perfect equilibrium. Conversely, if Γ is a game of perfect information, then every subgame-perfect equilibrium is also a sequential equilibrium.*

It is harder to show the following.

Proposition 2 *Let Γ be a finite extensive game of perfect recall. Then Γ admits a sequential equilibrium.*

Let us now return to the problem with *coalesce*. Here is an example. We note that (O, L_2) is part of a sequential equilibrium, with $\mu(x) = 1$. Indeed, consider the sequence $(\sigma_1^n(L_1), \sigma_1^n(R_1), \sigma_1^n(O)) = (n^{-1}, n^{-2}, 1 - n^{-1} - n^{-2})$. Because L_1 is "more likely" than R_1, the resulting probability assigned to x converges to 1. Sequential rationality is immediate.

Here is another normal-form with the same extensive form (an application of *COA* distinguishes these two extensive forms). Because the strategy profile

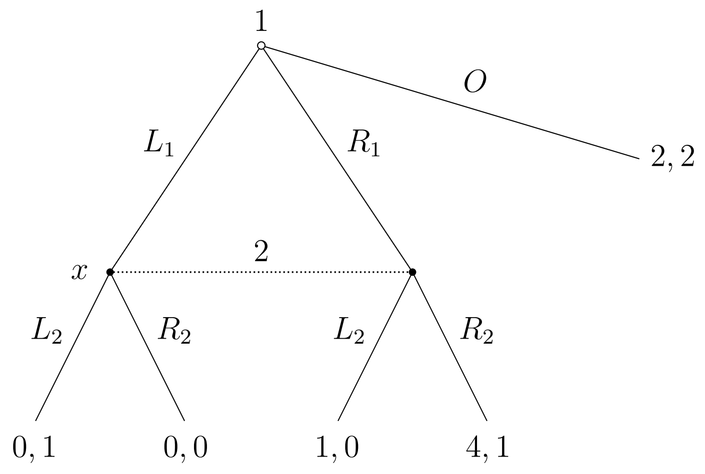


Figure 5: What are the sequential equilibria of this game?

in a sequential equilibrium must be an SPNE, we note that (R_1, R_2) must be specified in every sequential equilibrium, as it is the unique Nash equilibrium of the subgame starting after $\neg O$ (R_1 is strictly dominant in the subgame). Hence, sequential rationality requires player 1 to play $\neg O$.

So, we see that the set of sequential equilibrium outcomes is not independent of the extensive form that is considered: sequential equilibrium is not "invariant to transformations that leave the reduced normal form unchanged."

This example isn't as damning as the previous one, since, at least, the sets of sequential equilibria across those two games aren't disjoint. So perhaps we can aspire to a slightly less lofty goal: that the sets of sequential equilibria that share a common reduced normal form have non-empty intersection. (So, in particular, we give up on the transformation that allows us to focus on the fully reduced normal form.)

This is one of the few pieces of good news. This intersection is nonempty, and contains the set of *proper equilibria* of the normal form.

The inclusion can be strict, as the example in Figure 7 shows. (I leave it as an exercise: show that (A, V) is not proper, and is part of a sequential equilibrium in every extensive form representation with this normal form.)

Definition 17 *The Nash equilibrium σ is a proper equilibrium if there exists a sequence $(\sigma^n)_n \to \sigma$, with $\sigma^n \in \Sigma^0$, and $(\varepsilon^n)_n \to 0$, $\varepsilon^n > 0$, such that, for*

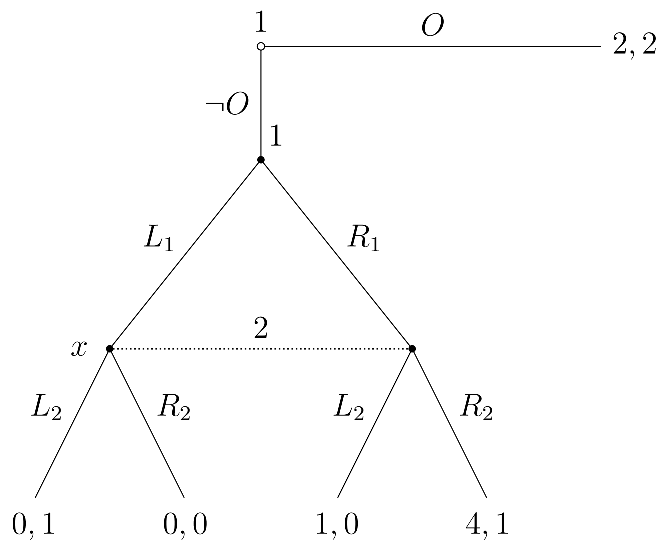


Figure 6: What are the sequential equilibria of this game?

	V	W	X	Y	Z
A	$1,1$	$3,1$	$3,1$	$0,0$	$1,1$
B	$1,1$	$2,0$	$2,0$	$0,0$	$1,2$
C	$1,1$	$1,0$	$2,1$	$1,0$	$1,0$
D	$1,1$	$2,1$	$1,0$	$1,0$	$1,0$

Figure 7: Proper vs. sequential equilibrium "in every extensive form."

every i, s_i, s_i',

$$u_i(s_i, \sigma_{-i}^n) < u_i(s_i', \sigma_{-i}^n) \Rightarrow \sigma_i^n(s_i) \leq \varepsilon^n \sigma_i^n(s_i').$$

This is a normal form concept. Two important properties of properness is:

1. Proper equilibria exist.
2. Every proper equilibrium induces a sequential equilibrium in every extensive form with the given normal form.

By "induce," we may consider the sequence $(\sigma^n)_n$ in the definition of properness to define the corresponding sequence of strategies in the sequential equilibrium (and derive the beliefs that support this strategy profile).

I skip the proof of existence, which is standard. As for the second part, let $\sigma = \lim_n \sigma^n$ as in the definition. Let $\tilde{\sigma}^n$ be the profile of (behavioral) strategies equivalent to σ^n (given Kuhn theorem), and let μ^n be the vector of conditional probabilities derived from $\tilde{\sigma}^n$ by Bayes rule at each information set. Let μ denote the limit (of a convergent subsequence) of μ^n, and similarly $\tilde{\sigma} := \lim_n \tilde{\sigma}^n$. We claim that (σ, μ) is a sequential equilibrium. Suppose not, that is, there is a player, i, and a (last) information set $h \in H_i$ such that $\tilde{\sigma}_i$ assigns positive probability to some action $a_i \in A(h)$ such that there exists $a_i' \in A(h)$ yielding a higher payoff conditional on h, given $\mu, \tilde{\sigma}_{-i}$, assuming that i follows $\tilde{\sigma}_i$ at all subsequent information sets. The same then holds for given $\mu^n, \tilde{\sigma}_{-i}^n$, for all n large.

It follows that every normal form strategy $\hat{\sigma}_i$ such that h has positive probability under $\hat{\sigma}_i$ given σ_{-i}^n, for n large enough, but chooses a_i at h, yields a lower payoff than the same strategy $\hat{\sigma}_i'$, except that it picks a_i' at h, and that it makes the same choices as $\tilde{\sigma}_i$ at any subsequent information set. Since σ is proper, σ_i^n must assign to $\hat{\sigma}_i$ a probability no more than ε^n times the probability it assigns to $\hat{\sigma}_i'$. Hence, it assigns to action a_i a probability at most $\varepsilon^n \times \#S_i$. Hence, taking $n \to \infty$, $\hat{\sigma}_i$ assigns probability zero to a_i, a contradiction.

So, to conclude, properness satisfies invariance, and captures sequential rationality. What's not to like?

2.2 Admissibility

Consider the following game. If B is not available (after all, it is strictly

	L	R
T	$1,1,1$	$0,0,1$
B	$0,0,1$	$0,0,1$

A

	L	R
T	$0,0,0$	$0,0,0$
B	$0,0,0$	$1,1,0$

B

Figure 8: Properness is sensitive to the addition of dominated strategies

dominated for player 3), the unique proper equilibrium is (T, L) (clearly, if P2 plays both actions with positive probability, P1 gets strictly more from T than B). But (B, R, A) is a proper equilibrium. To see this, let us use

$$\sigma_1^n(T) = \sigma_2^n(B) = (n+1)^{-2}, \quad \sigma_3^n(B) = (n+1)^{-1},$$

and so, for P1 again (clearly it's symmetric for P2, and P3 clearly takes an action that satisfies the requirement),

$$\begin{aligned} u_1(T, \sigma_{-1}^n) - u_1(B, \sigma_{-1}^n) &= \frac{1}{(n+1)^2}\left(1 - \frac{1}{n+1}\right) - \frac{1}{n+1}\left(1 - \frac{1}{(n+1)^2}\right) \\ &= -\frac{n}{(n+1)^2} < 0, \end{aligned}$$

and so indeed the definition of proper equilibrium is satisfied. Hence, properness is not "invariant to the addition of strictly dominated strategies."

This is an additional, desirable property we might want to require (in addition to invariance and sequential rationality); of course, we also want (i) existence, and (ii) the solution concept refines Nash equilibrium. In fact, we would like *admissibility*, namely, that the addition of *weakly* dominated strategies does not affect the set of predictions. While this might sound stronger, note that if we insist on invariance, sequential rationality, and invariance to strictly dominated strategies, admissibility follows: transform the extensive form game into one where the choice between the weakly dominated strategy and the one weakly dominating it is done at a last information set of that player, in the contingency where this choice matters. Sequential rationality then imposes that the weakly dominated strategy not be played.

The problem is that these requirements are inconsistent, as should be clear by now. We can have sequential rationality and invariance (proper equilibrium), but then we give up admissibility. We cannot have iterated dominance (so, iterated deletion of strictly dominated strategies) and admissibility too, as should be clear from the classic example in Figure 9. Strategy B is dominated

	L	R
T	$3,2$	$2,2$
M	$1,1$	$0,0$
B	$0,0$	$1,1$

Figure 9: Iterated Dominance

(in fact, strictly dominated) so by iterated dominance the solution should not change if B is deleted. But in the resulting game admissibility implies that (T, L) is the unique solution. Similarly, M is dominated and once it is deleted (T, R) is the unique solution.

So, either we give up on some, or we change the domain of solution concepts. Kohlberg and Mertens (1986) suggest to define solutions to be *set-valued*. While this sounds a bit far-fetched, think of sets of strategies corresponding to single equilibrium *outcomes*. After all, fixing an extensive form game, for any generic assignment of payoff vectors to the terminal nodes (an element of $\in \mathbf{R}^{n|Z|}$), there is a finite number of outcome distributions over Z induced by Nash equilibria.[2] (This is shown by Kreps and Wilson, 1982.) But a given equilibrium outcome often corresponds to sets of strategy profiles (for instance, actions at information sets off the equilibrium path are often not uniquely pinned down by the equilibrium outcome). So one might identify the solution of a game with an entire component (a connected set) of equilibria, mapped into a common equilibrium outcome.

Fix $\alpha \in (0, 1]^n$, and given $\sigma^0 \in \Sigma^0$, let $\delta_{\alpha,\sigma^0} : \Sigma \to \Sigma^0$ be defined by

$$\delta_{\alpha,\sigma^0}(\sigma) = \alpha\sigma^0 + (1 - \alpha)\sigma,$$

[2]A set is generic if its complement is a closed set of lower dimension.

Given the normal form of any given game $\Gamma := (T, P, H, C, p, u)$, we define the normal form of the game $\hat{\Gamma}_{\alpha,\sigma^0}$ by perturbing u according to δ_{α,σ^0}, namely, $\hat{\Gamma}_{\alpha,\sigma^0} = (T, P, H, C, p, \hat{u})$, where (in the normal form) $\hat{u}(\sigma) = u(\delta_{\alpha,\sigma^0}(\sigma))$. A closed subset $\tilde{\Sigma}$ of Nash equilibria of the normal form of Γ is *prestable* if, for every $\varepsilon > 0$, there exists $\varepsilon' > 0$ such that, for *any* $\alpha \in (0, \varepsilon']^n$, $\sigma^0 \in \Sigma^0$, then $\hat{\Gamma}_{\alpha,\sigma^0}$ has at least one Nash equilibrium within ε of $\tilde{\Sigma}$. A prestable set $\tilde{\Sigma}$ is *stable* if it contains no other prestable set.

It can be shown that any (finite) normal-form game admits at least one connected set of equilibria that contains a stable set $\tilde{\Sigma}$. Further, $\tilde{\Sigma}$ contains a stable set of any game obtained by eliminating any pure strategy that is weakly dominated, or that is not a best-reply to every every equilibrium in $\tilde{\Sigma}$.

2.3 Structural Consistency

Our earlier example (Figure 1 and 2) suggests that we have to give up on invariance (at least with respect to the fully reduced normal form): Sequential rationality is non-negotiable. Sequential equilibrium is an "upper solution concept." Of course, we saw already one drawback of sequential equilibrium, namely that the set of those depends on the extensive form. Here is another drawback, namely, that we cannot interpret the beliefs at off-path nodes as the "alternative explanation" a player might come up regarding what has happened. A belief system μ is *structurally consistent* if for each information set h, there is a strategy profile σ such that μ at h is derived from σ using Bayes' rule.

To understand why this is a problem, consider the game in Figure 10.

It is readily shown that in the unique sequential equilibrium of this game, Player 1 chooses R_1, Player 2 chooses R_2, and Player 3 must play both his actions with probability at least 1/3 (so, she must be indifferent). It follows that the belief μ she attaches to (x_3, y_3, w_3) must be $(0, 1/2, 1/2)$. But of course this belief cannot be derived from any (σ_1, σ_2) using Bayes' rule: independent choices would lead to a product distribution. The problem is that the set of consistent assessments is the closure of the set of such product distributions, and limits of product distributions aren't necessarily product distributions.

This isn't too much a problem, in our view: the belief system $(0, 1/2, 1/2)$ is not structurally consistent, but it is in the convex hull of structurally consistent

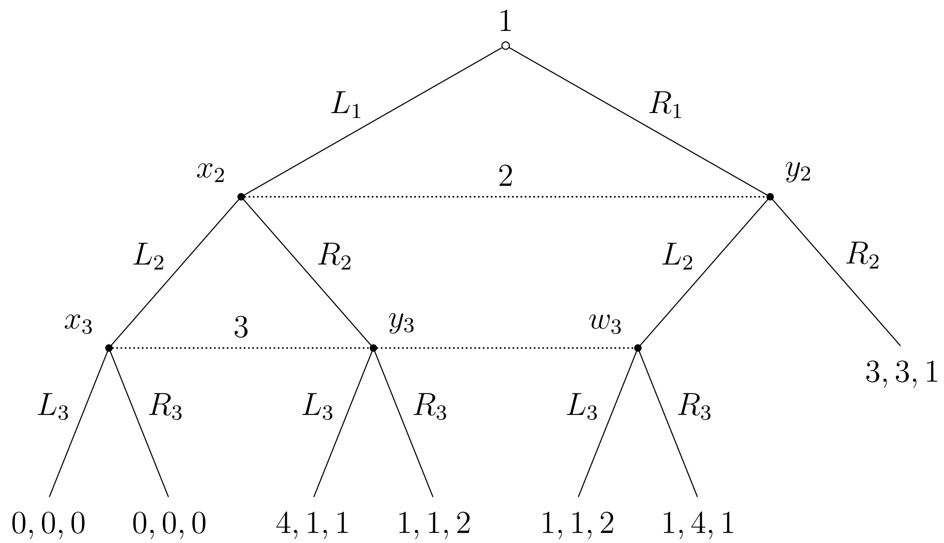


Figure 10: An example showing the difference between consistency and structural consistency.

beliefs (as the average of $(0, 0, 1)$ and $(0, 1, 0)$). Player 3 could arguably motivate her belief by being agnostic about which of the two deviations occurred, and assigning equal probability to either. More generally, it can be shown that any consistent belief is in the convex hull of the set of structurally consistent beliefs. (On the other hand, structural consistency does not imply consistency.)

2.4 Forward Induction

This doesn't imply that all sequential equilibria are equally reasonable. As mentioned, every solution should induce a solution in any subgame, but not every solution of a subgame has to be part of a solution of the game. Consider the game in Figure 11.

The assessment $\sigma = (R_1, R_2)$, $\mu(x_2) = 0$, is a sequential equilibrium. But C_1 is strictly dominated by R_1; whereas L_1 is not. In particular, if $\mu(x_2) = 1$, Player 2's optimal choice is L_2, which makes L_1 the optimal choice for Player

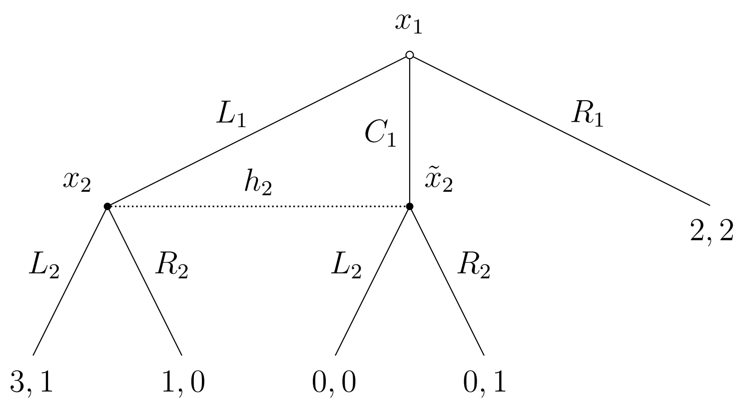


Figure 11: Forward Induction

1. Indeed, $\sigma = (L_1, L_2)$, $\mu(x_2) = 1$ is another sequential equilibrium that is selected by this logic. *Forward induction* does not assume that "bygones are bygones:" beliefs of a player at an information set, as well as what are plausible expectations regarding continuation play, should be informed by what other players might have given up for this information set to be reached. (Formally, forward induction is an order of deletion of weakly dominated strategies dictated by the extensive form.) Forward induction selects a unique sequential equilibrium in this game.

However, this logic, which also selects a unique sequential equilibrium in the game in Figure 12, might nonetheless be less compelling there. (Why?)

More problematic, forward induction might conflict with sequential rationality. Consider the game in Figure 13. Because l_1 is strictly dominated by R_1, whereas r_1 is not, Player 2 should logically infer from L_1 that Player 1 intends to play r_1 if she gets the opportunity to do so. Hence, Player 2 should pick R_2. Yet, the unique SPNE of this game is (R_1, L_2, l_1, l_2).

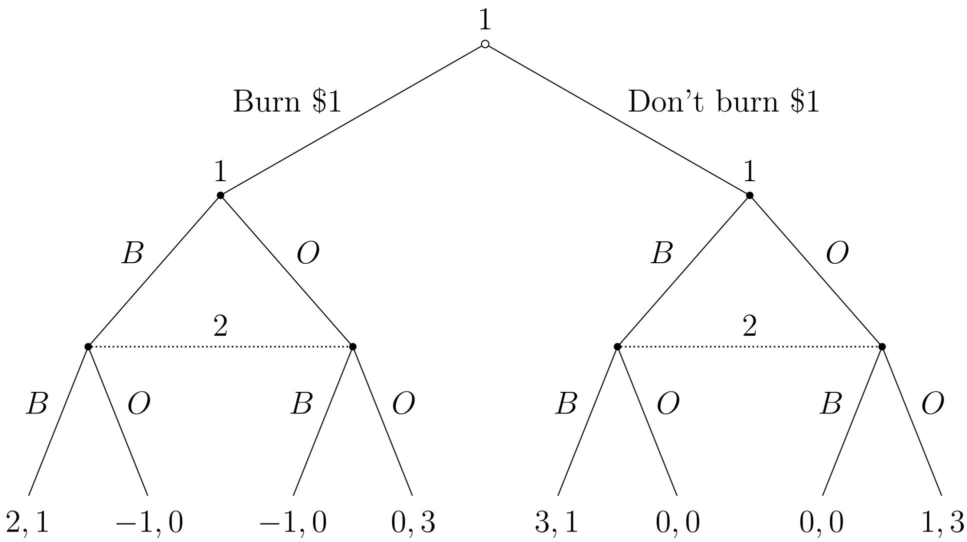


Figure 12: Extensive form of the money-burning game

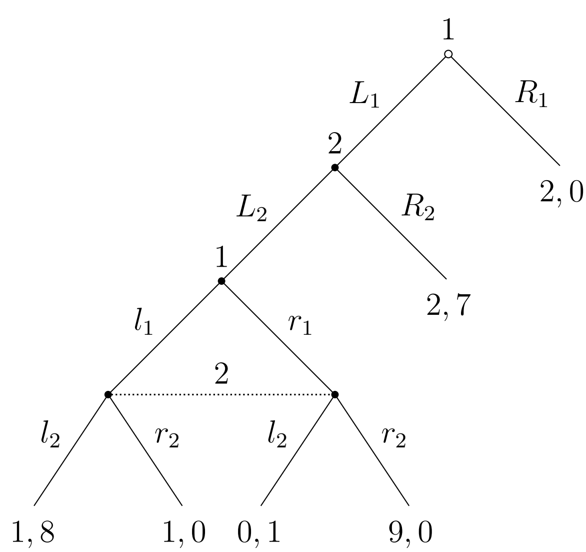


Figure 13: A game where backward and forward induction conflict